EE2006: Electrical Machines

Tutorial 5

1. A 200 V DC series motor runs at 700 rpm when operating at its full load current of 20 A. The motor resistance is 0.5 Ω and the magnetic circuit can be assumed unsaturated. What will be the speed if (i) the load torque is increased by 44% (ii) the motor current is 10 A.
[Ans: (i) 577.19 rpm (ii) 1436.84 rpm]

2. A 220 V DC series motor draws full load line current of 38 A at the rated speed of 600 rpm. The motor has armature resistance of 0.4 Ω and the series field resistance is 0.2 Ω. The brush voltage drop irrespective of load is 3.0 V, find:
(a) The speed of the motor when the load current drops to 19 A.
(b) The speed on removal of load when the motor takes only 1 A from supply.
(c) The internal horse power developed in each of the above cases.
Neglect the effect of armature reaction and saturation.
[Ans: (a) 1270.44 rpm (b) 25406.38 rpm (c) 5.236 hp, 0.29 hp]

3. A 440 V series motor has a total series resistance (including field) of 0.8 Ω. It takes 110 A to give the rated output at 800 rpm. Determine the resistance to be added to obtain rated torque:
(i) At starting (ii) At 600 rpm.
Assume that the flux is proportional to the current.
[Ans: (i) 3.2 Ω (ii) 0.8 Ω]

4. A separately excited DC generator has no-load voltage of 120 V at a field current of 2 A, when driven at 1500 rpm. Assuming that it is operating on the straight line portion of its saturation curve, calculate:
(a) The generated voltage when the field current is increased to 2.5 A.
(b) The generated voltage when the speed is reduced to 1400 rpm and the field current is increased to 2.84 A.
[Ans: (a) 150 V (b) 159.04 V]

5. A DC 60 kW shunt wound generator running at 500 rpm is supplying full load to a bus bar at 200 V. If the prime-mover input to the generator is removed and the machine now operates taking 5 kW from the bus bar, at what speed will it operate?
The armature and field resistances are 0.1 Ω and 100 Ω respectively.
Brush contact drop may be taken as 2 V. Neglect armature reaction.
[Ans: 421.4 rpm]

6. A DC shunt generator with an armature resistance of 0.75 Ω and a field resistance of 125 Ω supplies a load of 10 kW at a terminal voltage of 250 V.
Determine the efficiency of the generator assuming the iron losses, friction losses and windage losses amount to 600 W.
[Ans: 80.5%]

7. A separately excited DC generator is rated 6 kW, 120 V, 1200 rpm, and has the following open circuit characteristics at rated speed.
If (A) 0 0.1 0.2 0.3 0.4 0.5 0.6 0.8 1.0 1.2
Eg (V) 5 20 40 60 79 93 102 114 120 125
The machine parameters are Ra = 0.2 Ω and Rf = 100 Ω.
The machine is driven at 1200 rpm and the field current is adjusted at 0.8 A.
A load resistance, RL = 2 Ω is connected to the armature terminals.
Neglect armature reaction effect.
(a) Determine the quantity ke for the machine.
(b) Determine the generated voltage and the load current.
(c) Calculate the electrical power supplied to the load.
[Ans: (a) 0.9071 V/rad/s (b) 114 V (c) 5370.25 W]